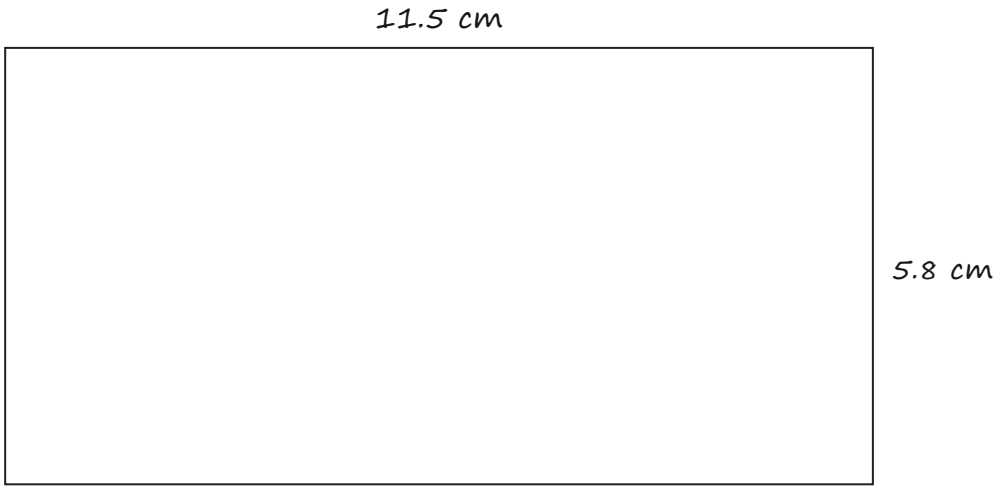
**12** The diagram shows a scale drawing of a tennis court.




The scale of the drawing is 1 : 200

Work out the perimeter of the real tennis court.
Give your answer in metres.

Measure on the original page: about 11.5 cm by 5.8 cm.

**1 mark**
Accurate measurement of a side

| | Plan | Real |
|---|---|---|
| | 1 | 200 |
| ×11.5 | | ×11.5 |
| | 11.5 | 2300 |

Both table columns are lengths in cm.

Length = 11.5 × 200 = 2300 cm

Width = 5.8 × 200 = 1160 cm

**1 mark**
Multiply the measured lengths by the scale factor

2300 cm = 23 m; 1160 cm = 11.6 m

**1 mark**
Divide centimetres by 100 to get metres

Perimeter = 2 × 23 + 2 × 11.6 = 69.2 m

**1 mark**
Add all four sides

**1 mark**
Answer within measurement tolerance

69.2 metres

**(Total for Question 12 is 5 marks)**